



SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

21CSC305P – Machine Learning

Unit 1: Complete Study Material & Reference Guide

Syllabus Coverage:

Introduction to Machine Learning (What & Why), Supervised & Unsupervised Learning, Polynomial Curve Fitting, Probability Theory, Discrete & Continuous Random Variables, Fundamental Rules of Probability, Bayes' Rule, Independence & Conditional Independence, Quantiles, Mean, Variance, Expectation, and Covariance.

Contents

1	Machine Learning: What and Why?	2
1.1	The Definition and Philosophy of Learning	2
1.2	The Pattern Recognition Framework	2
1.3	Why a Probabilistic Approach?	2
2	Supervised and Unsupervised Learning	2
2.1	Supervised (Predictive) Learning	3
2.2	Unsupervised (Descriptive) Learning	3
3	Polynomial Curve Fitting, Overfitting, and Regularization	3
3.1	The Polynomial Regression Model	3
3.2	The Sum-of-Squares Error Function	4
3.3	The Dilemma of Model Complexity: Underfitting vs. Overfitting	4
3.4	Weight Coefficient Explosion	4
3.5	Regularization (Ridge Regression)	4
4	Probability Theory: Discrete Random Variables and Fundamental Rules	5
4.1	Discrete Random Variables	5
4.2	The Two Fundamental Rules of Probability	5
4.3	Bayes' Rule	5
4.4	Independence and Conditional Independence	6
5	Continuous Random Variables and Probability Densities	7
5.1	Cumulative Distribution Function (CDF)	7
5.2	Probability Density Function (PDF)	7
5.3	Quantiles and Tail Area Probabilities	7
6	Expectation, Variance, and Covariance	8
6.1	Mathematical Expectation (Expected Value)	8
6.2	Variance and Standard Deviation	8
6.3	Covariance and Correlation	8

1 Machine Learning: What and Why?

1.1 The Definition and Philosophy of Learning

We are currently living in the **era of big data**. Trillions of web pages are indexed by search engines, billions of hours of video are uploaded to platforms daily, and corporations like Walmart process millions of transactions per hour, resulting in petabytes of data. This deluge of raw information calls for automated methods of data analysis, which is exactly what **Machine Learning (ML)** provides.

As defined by Kevin P. Murphy, machine learning is a set of methods that can automatically detect patterns in data and then use those uncovered patterns to predict future data or to perform other kinds of decision-making under uncertainty. Rather than writing static, hand-coded rules to solve a problem, machine learning stresses a **principled model-based approach**. By representing our assumptions and data using probability theory, we can systematically quantify uncertainty and perform optimal inference.

1.2 The Pattern Recognition Framework

A standard engineering pipeline for an automated pattern recognition and decision-making system consists of four fundamental, sequential steps:

1. **Sensing:** The physical acquisition of raw data from the environment via sensors (e.g., cameras, microphones, or pressure gauges).
2. **Preprocessing:** Cleaning and conditioning the raw data (e.g., noise reduction, normalization, contrast enhancement, or segmenting the region of interest from the background).
3. **Feature Extraction:** Transforming high-dimensional preprocessed data into a compact, informative set of numerical attributes (features) that represent the critical diagnostic information.
4. **Classification / Regression:** Evaluating the feature vector against a trained statistical model to output a discrete decision (class label) or a continuous predicted value.

1.3 Why a Probabilistic Approach?

Probability theory is the primary tool for representing and reasoning under uncertainty in machine learning. Uncertainty arises in many forms:

- **Sensor/Measurement Noise:** Physical measurements are inherently corrupted by noise.
- **Model Uncertainty:** We never know the true, exact underlying function that generated the data.
- **Incomplete Information:** We only observe a finite subset of variables, meaning some relevant causal factors are hidden (latent).

By adopting a probabilistic perspective, we represent features and target variables as random variables, and our learning algorithms act as mechanisms to estimate probability distributions, leading to highly robust and calibrated models.

2 Supervised and Unsupervised Learning

The two most dominant paradigms in machine learning are **supervised** and **unsupervised learning**, distinguished by the presence or absence of target training labels.

2.1 Supervised (Predictive) Learning

In the supervised learning approach, the goal is to learn a mapping from input features $\mathbf{x}$ to output targets y , given a labeled training dataset:

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$$

where $\mathbf{x}_i \in \mathbb{R}^D$ is a D -dimensional feature vector, and y_i is the corresponding label. The dataset $\mathcal{D}$ can be organized as an $N \times D$ **design matrix** $\mathbf{X}$, where each row is a feature vector, paired with a target vector $\mathbf{y}$ of length N . Depending on the nature of the target y , supervised learning is split into two categories:

1. **Classification:** The target y is a discrete categorical variable belonging to a finite set of classes, $y \in \{1, \dots, C\}$.
 - **Binary Classification:** $C = 2$ (e.g., Spam vs. Non-Spam, represented as $y \in \{0, 1\}$).
 - **Multiclass Classification:** $C > 2$, but each input belongs to exactly one class (e.g., classifying handwritten digits $y \in \{0, \dots, 9\}$).
 - **Multi-label Classification:** Each input can simultaneously belong to multiple classes (e.g., a photo containing both a dog and a cat).
2. **Regression:** The target y is a continuous real-valued scalar, $y \in \mathbb{R}$ (e.g., predicting global temperature anomalies or a student's final exam marks based on study hours).

2.2 Unsupervised (Descriptive) Learning

In unsupervised learning, we are given only input features without any target labels:

$$\mathcal{D} = \{\mathbf{x}_i\}_{i=1}^N$$

The goal is to discover "interesting structure" or "natural groupings" in the data, which is sometimes called **knowledge discovery**. Unsupervised learning can be formulated as a problem of **unconditional density estimation**, where we build joint probability models of the form $p(\mathbf{x} | \boldsymbol{\theta})$. This differs from the supervised case, which is a **conditional density estimation** problem of the form $p(y | \mathbf{x}, \boldsymbol{\theta})$. Common tasks in unsupervised learning include:

1. **Clustering:** Partitioning the dataset into groups of similar points (e.g., K-Means or Gaussian Mixture Models).
2. **Discovering Latent Factors (Dimensionality Reduction):** Reducing a high-dimensional feature space into a low-dimensional linear subspace (e.g., Principal Component Analysis) while preserving key structural variance.
3. **Discovering Graph Structure:** Learning conditional dependency relations between multiple variables to represent them as a sparse graphical model.

3 Polynomial Curve Fitting, Overfitting, and Regularization

3.1 The Polynomial Regression Model

To illustrate the core trade-offs of machine learning, consider the classic regression problem of fitting a polynomial curve to a set of N noisy observations. Suppose we have a continuous input variable x and a target variable t . The true generating function is a sinusoid, but our training data is corrupted by noise. We model the relationship using a polynomial function of order M :

$$y(x, \mathbf{w}) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M = \sum_{j=0}^M w_jx^j$$

where $\mathbf{w} = [w_0, w_1, \dots, w_M]^T$ is the weight parameter vector.

3.2 The Sum-of-Squares Error Function

To find the optimal weight parameters $\mathbf{w}^*$, we define an error function that measures the mismatch between our model's predictions $y(x_n, \mathbf{w})$ and the actual target training values t_n . The standard unregularized choice is the **sum-of-squares error function**:

$$E(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N \{y(x_n, \mathbf{w}) - t_n\}^2$$

Our objective is to find $\mathbf{w}^* = \arg \min_{\mathbf{w}} E(\mathbf{w})$, which can be solved analytically by taking the gradient with respect to $\mathbf{w}$ and setting it to zero (the Ordinary Least Squares solution).

3.3 The Dilemma of Model Complexity: Underfitting vs. Overfitting

The choice of the polynomial order M determines the capacity of our model:

- **Underfitting (M is too low, e.g., $M = 0$ or 1):** The model is too simple (a flat line or straight line) and cannot capture the true curved structure of the data, resulting in high training error and poor generalization.
- **Overfitting (M is too high, e.g., $M = 9$ with a small dataset $N = 10$):** The polynomial has too many degrees of freedom and oscillates wildly, tuning itself exactly to the random noise in the training set. This yields a training error of exactly zero ($E(\mathbf{w}^*) = 0$), but leads to extremely large, unstable generalization errors on unseen test data.

3.4 Weight Coefficient Explosion

A key indicator of overfitting is the behavior of the optimal weight coefficients $\mathbf{w}^*$. When a model overfits, the magnitudes of the coefficients **explode to extremely large values**. For example:

- For a simple fit ($M = 1$), w_1^* might be around -1.27 .
- For an overfitted fit ($M = 9$ with $N = 10$), the coefficients can reach magnitudes of hundreds of thousands (e.g., $w_6^* \approx -1,061,800$, while others are $+640,042$ to pass exactly through the noisy data points). These massive opposing values cancel each other out near the data points but cause wild, non-physical oscillations in between.

This can be mitigated by increasing the size of our dataset N . For a fixed complexity ($M = 9$), fitting to a larger dataset (e.g., $N = 100$) dampens the wild oscillations and stabilizes the weights.

3.5 Regularization (Ridge Regression)

To control overfitting when the dataset size is limited, we can apply **regularization** (also known as weight decay or Ridge regression). We add a quadratic penalty term to our error function to restrict the weights from growing too large:

$$\tilde{E}(\mathbf{w}) = E(\mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|^2 = \frac{1}{2} \sum_{n=1}^N \{y(x_n, \mathbf{w}) - t_n\}^2 + \frac{\lambda}{2} \sum_{j=0}^M w_j^2$$

where $\lambda \geq 0$ is the regularization parameter:

- If λ is very large (e.g., $\ln \lambda = 0$), the penalty dominates, forcing the weights to shrink near zero, leading to underfitting.
- If λ is extremely small (e.g., $\ln \lambda = -\infty$), the penalty has no effect, leading to unregularized overfitting.
- An optimal choice (such as $\ln \lambda = -18$ on standard benchmarks) successfully suppresses the weight magnitudes, yielding a smooth curve with excellent generalization.

4 Probability Theory: Discrete Random Variables and Fundamental Rules

4.1 Discrete Random Variables

A random variable X represents a numerical value associated with the outcome of a random event. A **discrete random variable** can take on values from a finite or countably infinite set. Its probability distribution is defined by a **probability mass function (PMF)**, denoted by $p(x) = P(X = x)$, which must satisfy:

$$0 \leq p(x) \leq 1 \quad \text{and} \quad \sum_{x \in X} p(x) = 1$$

4.2 The Two Fundamental Rules of Probability

Any complex multi-variable probabilistic model is built upon just two basic, non-negotiable mathematical axioms:

1. **The Sum Rule (Marginalization):** Given joint variables X and Y , the marginal probability $p(X)$ is obtained by summing the joint distribution over all possible states of Y :

$$p(X) = \sum_Y p(X, Y)$$

2. **The Product Rule:** The joint probability of X and Y is the product of the conditional probability of Y given X and the marginal probability of X :

$$p(X, Y) = p(Y | X)p(X)$$

4.3 Bayes' Rule

Starting from the symmetry of the product rule ($p(X, Y) = p(Y, X)$), we have:

$$p(Y | X)p(X) = p(X | Y)p(Y)$$

Dividing both sides by $p(X)$ yields the mathematical formulation of **Bayes' Rule**:

$$p(Y | X) = \frac{p(X | Y)p(Y)}{p(X)}$$

By applying the sum rule to the denominator, we express Bayes' rule as:

$$P(\omega_j | x) = \frac{p(x | \omega_j)P(\omega_j)}{p(x)} = \frac{p(x | \omega_j)P(\omega_j)}{\sum_{c=1}^C p(x | \omega_c)P(\omega_c)}$$

The Four Components of Bayes' Rule

- **Prior Probability** ($P(\omega_j)$): Our initial belief about the hypothesis/class ω_j before observing any data.
- **Likelihood** ($p(x | \omega_j)$): The probability of observing features x given that the true class is ω_j .
- **Evidence** ($p(x)$): The total probability of observing features x across all possible classes, acting as a normalization constant.
- **Posterior Probability** ($P(\omega_j | x)$): Our updated belief about the class membership after incorporating the observed features.

Bayes' Rule in Medical Diagnostics (Base Rate Fallacy)

Problem Statement: A patient tests positive on a diagnostic test that is 99% accurate. The disease has a baseline prior rate of 1 in 10,000 in the general population. Calculate the actual probability that the patient has the disease.

Variables: Let $y = 1$ (has disease), $y = 0$ (healthy), $x = 1$ (tests positive).

$$\begin{aligned}
 P(y = 1) &= 0.0001 & \implies & P(y = 0) = 0.9999 \\
 P(x = 1 | y = 1) &= 0.99 & \text{(Sensitivity)} & \\
 P(x = 0 | y = 0) &= 0.99 & \implies & P(x = 1 | y = 0) = 0.01 \quad \text{(False Positive Rate)}
 \end{aligned}$$

Applying Bayes' Theorem:

$$\begin{aligned}
 P(y = 1 | x = 1) &= \frac{P(x = 1 | y = 1)P(y = 1)}{P(x = 1 | y = 1)P(y = 1) + P(x = 1 | y = 0)P(y = 0)} \\
 &= \frac{0.99 \times 0.0001}{(0.99 \times 0.0001) + (0.01 \times 0.9999)} \\
 &= \frac{0.000099}{0.000099 + 0.009999} = \frac{0.000099}{0.010098} \approx \mathbf{0.0098 \text{ (0.98\%)}}
 \end{aligned}$$

Conclusion: Despite a 99% accurate test, the actual chance of having the disease is less than 1% because the prior probability is extremely low.

4.4 Independence and Conditional Independence

1. **Unconditional Independence:** Two random variables X and Y are independent (denoted $X \perp Y$) if knowledge of one provides no information about the other:

$$p(X, Y) = p(X)p(Y) \iff p(Y | X) = p(Y)$$

Assuming independence significantly simplifies a model. A general joint distribution on two discrete variables with 6 and 5 states requires $(6 \times 5) - 1 = 29$ parameters. Assuming independence reduces this to $(6 - 1) + (5 - 1) = 9$ parameters.

2. **Conditional Independence (CI):** Two variables X and Y are conditionally independent given a third variable Z (denoted $X \perp Y | Z$) if their dependency is completely mediated by Z :

$$p(X, Y | Z) = p(X | Z)p(Y | Z) \iff p(X | Y, Z) = p(X | Z)$$

Conditional independence is the foundational assumption of Bayesian Networks (like Naive Bayes classifiers), allowing us to represent highly complex joint distributions as modular, factorized pieces.

5 Continuous Random Variables and Probability Densities

5.1 Cumulative Distribution Function (CDF)

A **continuous random variable** can take on any real-valued scalar within a continuous interval. Its distribution is defined by its **Cumulative Distribution Function (CDF)**:

$$F(q) = P(X \leq q)$$

The CDF is a monotonically increasing function mapping to the interval $[0, 1]$.

5.2 Probability Density Function (PDF)

The **Probability Density Function (PDF)**, denoted by $f(x)$ or $p(x)$, is defined as the derivative of the CDF:

$$f(x) = \frac{d}{dx}F(x)$$

The probability that a continuous variable X falls within a finite interval $[a, b]$ is given by the integral of the PDF over that range:

$$P(a < X \leq b) = \int_a^b f(x)dx = F(b) - F(a)$$

A valid PDF must satisfy:

$$f(x) \geq 0 \quad \text{for all } x \quad \text{and} \quad \int_{-\infty}^{\infty} f(x)dx = 1$$

Note that unlike discrete probabilities, a probability density $f(x)$ can exceed 1.0 (e.g., a uniform PDF over $[0, 0.5]$ has density $f(x) = 2.0$), as long as the total integrated area remains exactly 1.0.

5.3 Quantiles and Tail Area Probabilities

The α -quantile of a distribution is the value x_α such that the probability of X being less than or equal to x_α is exactly α :

$$F(x_\alpha) = \alpha \quad \Longleftrightarrow \quad x_\alpha = F^{-1}(\alpha)$$

- **The Median:** $F^{-1}(0.5)$ divides the probability mass equally in half.
- **The Interquartile Range:** $Q_1 = F^{-1}(0.25)$ (25th percentile) and $Q_3 = F^{-1}(0.75)$ (75th percentile).
- **95% Confidence Interval:** For a standard Gaussian distribution $\mathcal{N}(0, 1)$, the central 95% mass is bounded by the quantiles $[\Phi^{-1}(0.025), \Phi^{-1}(0.975)] \approx [-1.96, 1.96]$. For a general Gaussian $\mathcal{N}(\mu, \sigma^2)$, this corresponds to the range $\mu \pm 1.96\sigma$, which is often approximated as $\mu \pm 2\sigma$.

Continuous PDF Normalization and Probability

Problem Statement: A continuous random variable X has the raw density $f(x) = x^2$ over the domain $x \in [0, 1]$. Normalize this PDF and compute the probability that X takes a value in $[0.5, 1]$.

Step 1: Normalizing the PDF. We introduce a scaling constant k such that:

$$\int_0^1 kx^2 dx = 1 \implies k \left[\frac{x^3}{3} \right]_0^1 = 1 \implies k \left(\frac{1}{3} - 0 \right) = 1 \implies \mathbf{k = 3}$$

The normalized PDF is $f(x) = 3x^2$ for $x \in [0, 1]$.

Step 2: Calculating the Probability.

$$P(0.5 \leq X \leq 1) = \int_{0.5}^1 3x^2 dx = [x^3]_{0.5}^1 = 1^3 - (0.5)^3 = 1 - 0.125 = \mathbf{0.875 \text{ (87.5\%)}}$$

6 Expectation, Variance, and Covariance

6.1 Mathematical Expectation (Expected Value)

The expectation (or mean) of a random variable representing the "center of gravity" of its distribution is defined as:

$$\mathbb{E}[X] \equiv \mu = \sum_{x \in X} x \cdot p(x) \quad (\text{Discrete}) \quad \text{or} \quad \int_{-\infty}^{\infty} x \cdot f(x) dx \quad (\text{Continuous})$$

A critical property of expectation is its **linearity**:

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c$$

for any constants a, b, c , regardless of whether X and Y are independent.

6.2 Variance and Standard Deviation

The variance measures the "spread" or uncertainty of a distribution around its mean:

$$\text{Var}[X] \equiv \sigma^2 = \mathbb{E}[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

Expanding this quadratic expectation yields a highly useful algebraic shortcut:

$$\text{Var}[X] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[\mathbf{X}^2] - (\mathbb{E}[\mathbf{X}])^2$$

The standard deviation is simply the square root of the variance, $\sigma = \sqrt{\text{Var}[X]}$, representing spread in the same units as the random variable itself.

6.3 Covariance and Correlation

The covariance measures the degree to which two random variables X and Y are linearly related:

$$\text{cov}[X, Y] \equiv \sigma_{XY} = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Positive Covariance: Large values of X tend to occur with large values of Y .

Negative Covariance: Large values of X tend to occur with small values of Y .

Independence relationship: If X and Y are independent, then $\text{cov}[X, Y] = 0$. However, a covariance of zero does **not** imply independence in a general setting, unless the variables are jointly Normally distributed. Non-linear dependencies can yield a covariance of exactly zero.

Variance of a Sum:

$$\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y] + 2\text{cov}[X, Y]$$

If X and Y are independent, this simplifies to the additive property: $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$.

Correlation: To normalize covariance into a scale-free metric between $[-1, 1]$, we divide by the product of their standard deviations:

$$\rho_{X,Y} = \text{corr}[X, Y] = \frac{\text{cov}[X, Y]}{\sigma_X \sigma_Y}$$

Discrete Expectation

Problem Statement: Two discrete variables are defined by the joint distribution $f(x, y)$ below. Calculate the expected values μ_X , μ_Y , and the covariance $\text{cov}(X, Y)$.

$f(x, y)$	$X = 5$	$X = 6$	$X = 7$	Marginal $f_Y(y)$
$Y = 8$	0.0	0.4	0.1	0.5
$Y = 9$	0.3	0.0	0.2	0.5
Marginal $f_X(x)$	0.3	0.4	0.3	1.0

Step 1: Expected Values.

$$\mu_X = \sum_x x f_X(x) = 5(0.3) + 6(0.4) + 7(0.3) = 1.5 + 2.4 + 2.1 = \mathbf{6.0}$$

$$\mu_Y = \sum_y y f_Y(y) = 8(0.5) + 9(0.5) = 4.0 + 4.5 = \mathbf{8.5}$$

Step 2: Covariance.

$$\begin{aligned}
 \text{cov}(X, Y) &= \sum_x \sum_y f(x, y)(x - \mu_X)(y - \mu_Y) \\
 &= 0(5 - 6)(8 - 8.5) + 0.4(6 - 6)(8 - 8.5) + 0.1(7 - 6)(8 - 8.5) \\
 &\quad + 0.3(5 - 6)(9 - 8.5) + 0(6 - 6)(9 - 8.5) + 0.2(7 - 6)(9 - 8.5) \\
 &= 0 + 0 + 0.1(1)(-0.5) + 0.3(-1)(0.5) + 0 + 0.2(1)(0.5) \\
 &= -0.05 - 0.15 + 0.10 = \mathbf{-0.10}
 \end{aligned}$$

Continuous Expected Value and Variance Calculations

Problem Statement: For a continuous random variable X defined over the domain $x \in [1, 3]$ with the normalized PDF $f(x) = \frac{3}{26}x^2$, calculate its expected value and variance.

Step 1: Expected Value $\mathbb{E}[X]$.

$$\mathbb{E}[X] = \int_1^3 x \cdot \left(\frac{3}{26}x^2\right) dx = \frac{3}{26} \left[\frac{x^4}{4}\right]_1^3 = \frac{3}{104}(3^4 - 1^4) = \frac{3}{104}(80) = \frac{30}{13} \approx 2.308$$

Step 2: Expected Value of Squared Variable $\mathbb{E}[X^2]$.

$$\mathbb{E}[X^2] = \int_1^3 x^2 \cdot \left(\frac{3}{26}x^2\right) dx = \frac{3}{26} \left[\frac{x^5}{5}\right]_1^3 = \frac{3}{130}(3^5 - 1^5) = \frac{3}{130}(242) = \frac{363}{65} \approx 5.585$$

Step 3: Variance $\text{Var}(X)$.

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{363}{65} - \left(\frac{30}{13}\right)^2 = \frac{363}{65} - \frac{900}{169} = \frac{4719 - 4500}{845} = \frac{219}{845} \approx 0.259$$